

(ANSWER)
Physical quantities (Specified by (1) number and (2) unit)

Example1.

Suggest two different units for each of the following physical quantities

(a) Length cm / m / km / inch / mile(b) volume cm³ / m³ / dm³(c) Time s / min / hr / yr(d) speed m/s, km/h

Fundamental quantity	Base SI unit	Symbol for the unit
1. Length	metre	m
2. Mass	kilogram	kg
3. Time	second	s
4. Temperature	Kelvin / degree of Celsius	K / °C
5. Electric current	ampere	A

Remarks: unit for other physical quantities can be expressed in terms of above 5 base or other SI units)

1. Give the SI unit for each of the following in terms of the base units

(a) $\text{Density} = \frac{\text{mass}}{\text{volume}}$

(b) $\text{Speed} = \frac{\text{distance}}{\text{time}}$

(c) $\text{Rate of temperature change} = \frac{\text{temperature change}}{\text{time}}$

(d) $\text{Rate of change of speed} = \frac{\text{speed}}{\text{time}}$

(e) $\text{Energy} = \text{power} \times \text{time}$

(a)	<u>kg / m³</u>
(b)	<u>m / s</u>
(c)	<u>°C / s</u>
(d)	<u>ms⁻¹ / s</u>
(e)	<u>W s, J</u>

(ANSWER)**Power of ten notation** (Scientific notation)

(1) Very large or very small numbers can be expressed more neatly and conveniently in a 'shorthand' form using powers of ten instead of writing many zeros

$$1\,000\,000 = 10^6$$

$$0.1 = \frac{1}{10} = \frac{1}{10^1} = 10^{-1}$$

$$100\,000 = 10^5$$

$$0.01 = \frac{1}{100} = \frac{1}{10^2} = 10^{-2}$$

$$10\,000 = 10^4$$

$$0.001 = \frac{1}{1\,000} = \frac{1}{10^3} = 10^{-3}$$

$$1\,000 = 10^3$$

$$0.0001 = \frac{1}{10\,000} = \frac{1}{10^4} = 10^{-4}$$

(2) The index superscript on the top right corner of a 10 is called the power of 10. Note that 1 can be written as 10^0 . For numbers less than 1, the indices are negative

For examples:

$$400 = 4 \times 100 = 4 \times 10^2$$

$$30\,000 = 3 \times 10\,000 = 3 \times 10^4$$

$$7\,200\,000 = 7.2 \times 1\,000\,000 = 7.2 \times 10^6$$

$$0.05 = 5 \times 0.01 = 5 \times 10^{-2}$$

$$0.000\,02 = 2 \times 0.000\,01 = 2 \times 10^{-5}$$

$$0.000\,000\,064 = 6.4 \times 0.000\,000\,01 = 6.4 \times 10^{-8}$$

2. Write the following in power-of-ten form with 1 figure before the decimal point

(a) 4600 4.6×10^3

(b) 2220000 2.22×10^6

(c) 31800000 3.18×10^7

(d) 0.0008 8×10^{-4}

(e) 0.000415 4.15×10^{-4}

(f) 0.00000606 6.06×10^{-6}

3. Write out the following in full

(a) 4×10^4 40000

(b) 8.54×10^6 8540000

(c) 7×10^{-4} 0.0007

(d) 3.5×10^{-7} 0.00000035

(ANSWER)**Prefixes**

- (1) When units of physical quantities are found to be too large or too small, multiples or sub-multiples are used. These usually go up or go down by factor of 1000 (sometimes by 10 or 100)

Multiple	Prefix	Symbol
1 000 (10 ³)	kilo	k [千]
1 000 000 (10 ⁶)	mega	M [兆]

Sub-multiple	Prefix	Symbol
0.1	deci	d [分]
0.01	centi	c [厘]
0.001 (10 ⁻³)	milli	m [毫]
0.000 001 (10 ⁻⁶)	micro	μ [微]

For examples:

1 000 m	= 10 ³ m	= 1 km (kilometre)
1 000 000 V	= 10 ⁶ V	= 1 MV (megavolt)
0.01 m	= 10 ⁻² m	= 1 cm (centimetre)
0.001 g	= 10 ⁻³ g	= 1 mg (milligram)
0.000 001 s	= 10 ⁻⁶ s	= 1 μs (microsecond)

- (2) The prefixes should be used so that, usually, the numerical part will lie between 0.1 and 1000 (The rule is not rigid)

For examples: 25 000 000 V is preferred to write as 25 MV and NOT 25 000 kV.

0.000 05 s is preferred to write as 50 μs and NOT 0.05 ms.

4. Write the following using the most suitable prefixes

(a) 0.007 s 7 ms

(b) 3000 g 3 kg

(c) 4000000 W 4 MW

(d) 90000 m 90 km

(e) 30000 V 30 kV

(f) 0.000085 m 85 μm

(g) 2.4 × 10³ A 2.4 kA

(h) 7 × 10⁴ g 70 kg

(i) 4.8 × 10⁻³ s 4.8 ms

(j) 0.4 × 10⁻⁶ W 0.4 μW

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5. Express the following values using base units (i.e. without prefixes) and write your answer in power-of-ten form

(a) $122 \mu\text{m}$ $\underline{1.22 \times 10^{-4} \text{ m}}$

(b) 65 kV $\underline{6.5 \times 10^4 \text{ V}}$

(c) 0.03 mA $\underline{3 \times 10^{-5} \text{ A}}$

(d) 0.3 kg $\underline{300 \text{ g}}$ $\underline{= 3 \times 10^2 \text{ g}}$

(e) 3000 MW $\underline{3 \times 10^9 \text{ W}}$

(f) 0.000085 km $\underline{0.085 \text{ m}}$ $\underline{= 8.5 \times 10^{-2} \text{ m}}$

6. The speed of a car can be expressed as kilometres per hour (km/h).

(a) Convert 36 km/h to m/s.

(b) Convert 50 km/h to m/s.

(c) Convert 100 km/h to m/s.

(a)
$$\frac{36 \times 1000 \text{ m}}{3600 \text{ s}} = 10 \text{ m/s} \quad (\text{ms}^{-1})$$

(b)
$$\frac{50 \times 1000}{3600} = 13.9 \text{ m/s}$$

(c)
$$\frac{100 \times 1000}{3600} = 27.8 \text{ m/s}$$

7. Express the following in base SI units

(a) 72 km/h

(b) $0.002 \text{ kJg}^{-1}\text{C}^{-1}$

(b)
$$\frac{0.002 \text{ kJ}}{(1\text{g})(1^\circ\text{C})} = \frac{0.002(1000)\text{J}}{(\frac{1}{1000}\text{kg})(1^\circ)} = 2000 \text{ Jkg}^{-1}\text{C}^{-1}$$

8. Express 1 cm^2 in its base SI unit and write it in power-of-ten form

Express 1 cm^2 in its base SI unit and write it in power-of-ten form

$\Rightarrow (1\text{cm})(1\text{cm})$

$$\frac{0.01 \text{ m} \times 0.01 \text{ m}}{= 0.0001 \text{ m}^2}$$

$$= 1 \times 10^{-4} \text{ m}^2$$